NYTU ANALYTICS

Business Proposal

# VENDOR RISK ANALYSIS & PROCUREMENT INSIGHTS SOLUTION

**DATA-DRIVEN VENDOR RISK MANAGEMENT & PROCUREMENT OPTIMIZATION**

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| **Team:** | \| | **NYTU Analytics** |
| **Course:** | \| | INFO-465-002 \| Projects in Information Systems |
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**EXECUTIVE SUMMARY**

NYTU Analytics has developed a scalable, data-driven solution to help organizations gain visibility into vendor risk, spending concentration, and supplier diversity. By leveraging KNIME for automated data processing and Tableau for interactive visualization, this solution transforms raw procurement data into actionable strategic insights, enabling smarter vendor management, reduced operational risk, and more inclusive procurement.

## 1. Problem Statement

Organizations often lack clear visibility into vendor risk, spending concentration, and supplier diversity. Without structured analysis, it becomes difficult to identify high-risk vendors, reduce dependency, and ensure inclusive procurement strategies. This can lead to:

- Operational disruptions from over-reliance on key vendors
- Inefficient decision-making driven by incomplete data
- Missed opportunities for diversification and inclusive procurement

## 2. Proposed Solution

NYTU Analytics developed a data-driven solution that analyzes procurement data to identify vendor risk, spending concentration, and supplier diversity. The solution leverages KNIME for data preparation and risk modeling, and Tableau for visualization and decision support.

**The system transforms raw procurement data into actionable insights by:**

- Cleaning and standardizing vendor data for consistency
- Aggregating total spend at the vendor level
- Classifying vendors into risk categories based on spending concentration
- Visualizing key insights through an interactive Tableau dashboard

## 3. Methodology

### Data Preparation

Procurement data from 2024 and 2025 were combined and cleaned to ensure consistency across all vendor records. This process included standardizing naming conventions and preparing the unified dataset for downstream analysis.

### Data Processing & Risk Classification

Using KNIME, we built a structured, automated workflow to:

- Aggregate vendor-level spend across the combined dataset
- Apply classification rules to assign vendors to High, Medium, or Low risk tiers
- Generate a clean, analysis-ready output dataset

This workflow is automated and scalable, designed to be reused seamlessly with future procurement data.

### Data Visualization

A Tableau dashboard was developed to present findings in an accessible, interactive format, including:

- Top vendors by spend with associated risk classification
- Spend concentration analysis using a Pareto framework
- Total procurement spend summary across the full dataset
- SWAM vs. Non-SWAM vendor spending comparison

## 4. Key Insights

| 60% | 14 | HIGH |
|---|---|---|
| **SPEND CONCENTRATION** among just 14 vendors | **KEY VENDORS** drive majority of spend | **RISK EXPOSURE** among top-spend vendors |

These findings highlight potential exposure to vendor-related disruptions and emphasize the importance of managing vendor relationships strategically. High-spend vendors classified as high-risk represent the most critical area for immediate action.

- A small number of vendors account for a large portion of total spend
- Approximately 60% of the total spend is concentrated among just 14 vendors
- Several high-spend vendors are classified as high risk, indicating significant dependency
- SWAM vendors represent a smaller portion of overall spending compared to non-SWAM vendors

> *Note: The SWAM analysis is based on the subset of data where SWAM classification was available.*

## 5. Strategic Recommendations

Based on analysis of the procurement dataset, NYTU Analytics recommends the following actions:

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| 1 | **Diversify Vendor Base** | Reduce dependency on the small number of vendors currently accounting for the majority of spend. Introduce alternative suppliers to minimize concentration risk. |
| 2 | **Monitor High-Risk Vendors** | Establish enhanced monitoring protocols for high-spend, high-risk vendors. Proactive oversight reduces the likelihood and impact of vendor-related disruptions. |
| 3 | **Increase SWAM Participation** | Expand engagement with Small, Women-owned, and Minority-owned (SWAM) vendors to improve supplier diversity and meet inclusive procurement goals. |
| 4 | **Implement Ongoing Analytics** | Leverage the reusable KNIME workflow to continuously track vendor risk as new procurement data becomes available, enabling proactive decision-making over time. |

## 6. Business Impact

This solution delivers measurable value across key organizational dimensions:

| Risk Reduction | Better Decisions | Supplier Diversity | Resilient Strategy |
|---|---|---|---|
| Reduces operational risk from vendor concentration and dependency | Improves procurement decisions through real-time, data-driven insights | Increases SWAM vendor participation for more inclusive procurement | Supports a more resilient and scalable procurement strategy |

## 7. Technical Value

From a technical standpoint, this solution delivers both immediate and long-term value:

- Utilizes KNIME to automate data cleaning, transformation, and risk classification, eliminating manual effort and reducing errors
- Creates a scalable, reusable workflow that can be refreshed with new procurement data each cycle
- Integrates Tableau for clear, interactive visualization of vendor risk and spend insights
- Produces a clean, structured output dataset suitable for further analysis or integration into enterprise systems

## 8. Conclusion

> Our solution bridges the gap between raw procurement data and strategic decision-making. By identifying vendor risk, spending concentration, and supplier diversity

> opportunities, the organization is better equipped to reduce dependency, improve efficiency, and make more informed procurement decisions.
>
> This approach not only addresses current challenges but also provides a scalable, automated framework for ongoing procurement analytics, delivering sustained value well beyond the initial deployment.

*— NYTU Analytics | INFO-465-002 | Spring 2026 —*